
Laser power consumption of soliton formation in a bidirectional Kerr resonator

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S1. Simulated comb spectra with varied detuning

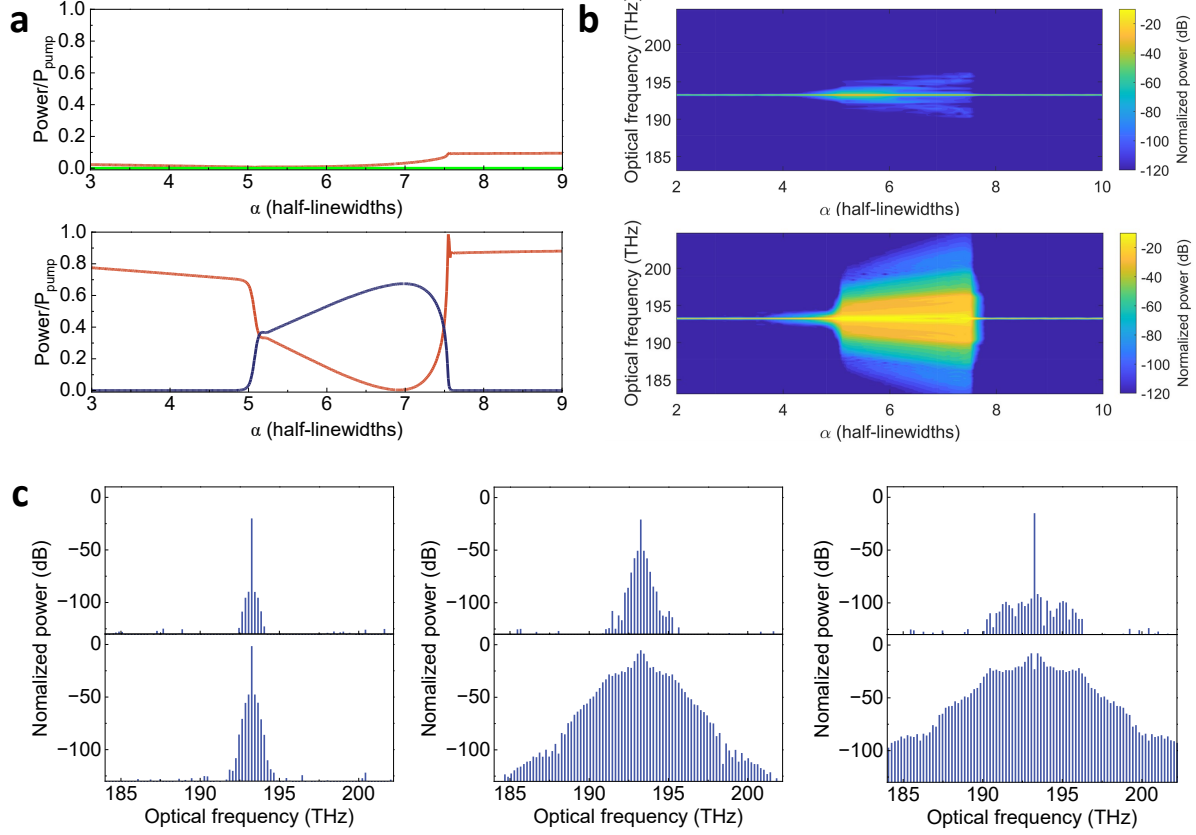


Fig. S1. Simulated soliton formation process. The upper and lower sub-panels represent the forward and backward output, respectively. **a**, The pump power (red trace) and comb power (green and blue traces) as α increases. **b**, Simulated evolution of the comb spectra as a function of α . **c**, Comb spectra in the case with $\alpha=4.7$ (left), 5.35 (middle), and 6.98 (right) half-linewidths, respectively.

Figure S1 shows the simulated soliton formation process corresponding to Fig. 1d, with the upper and lower sub-panels representing the forward and backward outputs, respectively. In Fig. S1a, we plot the pump power (red trace) and comb power (green and blue traces) as a function of α . The simulated backward comb power is much higher than that in the forward direction, which is consistent with our observation in the experiment: the CE of backward comb is below 0.5%. Therefore, we can ignore the forward comb power when calculating CE. The backward comb power, shown by the blue trace, varies smoothly as α increases, indicating that there is no breathing soliton state. Figure S1b presents the evolution of the comb spectrum with varied α , while Fig. S1c shows the comb spectra corresponding to α values of 4.7 (left), 5.35 (middle), and 6.98 (right) half-linewidths, respectively. Here the comb line power is normalized

to the pump power at the input of the PhCR chip. In the case with $\alpha = 6.98$ half-linewidths, the pump is highly depleted in the backward direction and the corresponding CE is 67% in the simulation.

S2. Simulated intensity profile and comb spectra with varied ϕ

Figure S2 presents the intensity profile and comb spectra predicted by our LLE simulation. We plot the pulse intensity as a function of azimuthal angle along the resonator's circumference and vary ϕ from 0 to 2π with a step size of 0.25π . It should be noted that the step size of ϕ in Fig. 1e is much smaller to show a more accurate trace of CE versus ϕ , while other parameters are the same in Fig. 1e and Fig. S2: normalized pump power $F^2 = 4.41$, $\epsilon_{\text{PhC}} = 6$, FSR = 200 GHz, $K = 3$, $Q_i = 2.7 \times 10^6$, and $D_2 = -2\pi \times 8.5$ MHz. In the case with $\phi = 1.0\pi$, our nanophotonic circuit cannot generate any comb due to the large threshold power that results from the destructive interference between the reflected pump by the reflector and the backward pump in the PhCR. We also observe that the comb line corresponding to the pump frequency is lower than the adjacent comb lines for ϕ of 0π (2π), 0.25π , 0.5π , and 1.75π , indicating that our nanophotonic circuit can maintain a high conversion efficiency over a wide range of ϕ . This feature is important for mass production of the high efficiency devices since fabrication uncertainties may lead to fluctuations in ϕ achieved in the experiment.

S3. Measured comb spectra with varied ϕ

In the experiment, we design multiple devices with varied ϕ to investigate the dependence of CE on ϕ . We sweep ϕ by varying the distance between the PhCR and the reflector or simply rotating the PhCR. For each set of PhCR parameters, we design eight devices to sweep ϕ with a step of 0.25π . Figure S3 presents the comb spectra corresponding to the measured CE (circles) in Fig. 1e. The device for $\phi = 0.25\pi$ has a defect from fabrication (broken waveguide) and the threshold power in the case with $\phi = 1.0\pi$ is too large for Kerr comb generation. Therefore, we only plot comb spectra corresponding to other values of ϕ in Fig. S3. In the case with $\phi = 0\pi$ and 1.75π , the pump is highly depleted, while in the case with $\phi = 0.75\pi$ and 1.25π , the residual pump power is significantly higher than other comb lines. This trend agrees with simulation results in Fig. S2, indicating that ϕ plays an important role in determining the CE of our nanophotonic circuit.

S4. Simulated stability chart for the nanophotonic circuit

We plot the simulated stability chart with varied comb detuning and normalized pump power; see Fig. S4. Here we use the same definition of comb detuning as that in [1]. We set $K = 10$, $\epsilon_{\text{PhC}} = 12$, and $\phi = 0\pi$, while other parameters are the same as those in Fig. S2. Figure S4a shows the comb states predicted by our simulation, with different color indicating different comb states. In Fig. S4b, we plot single-soliton CE with a color bar on a logarithmic scale,

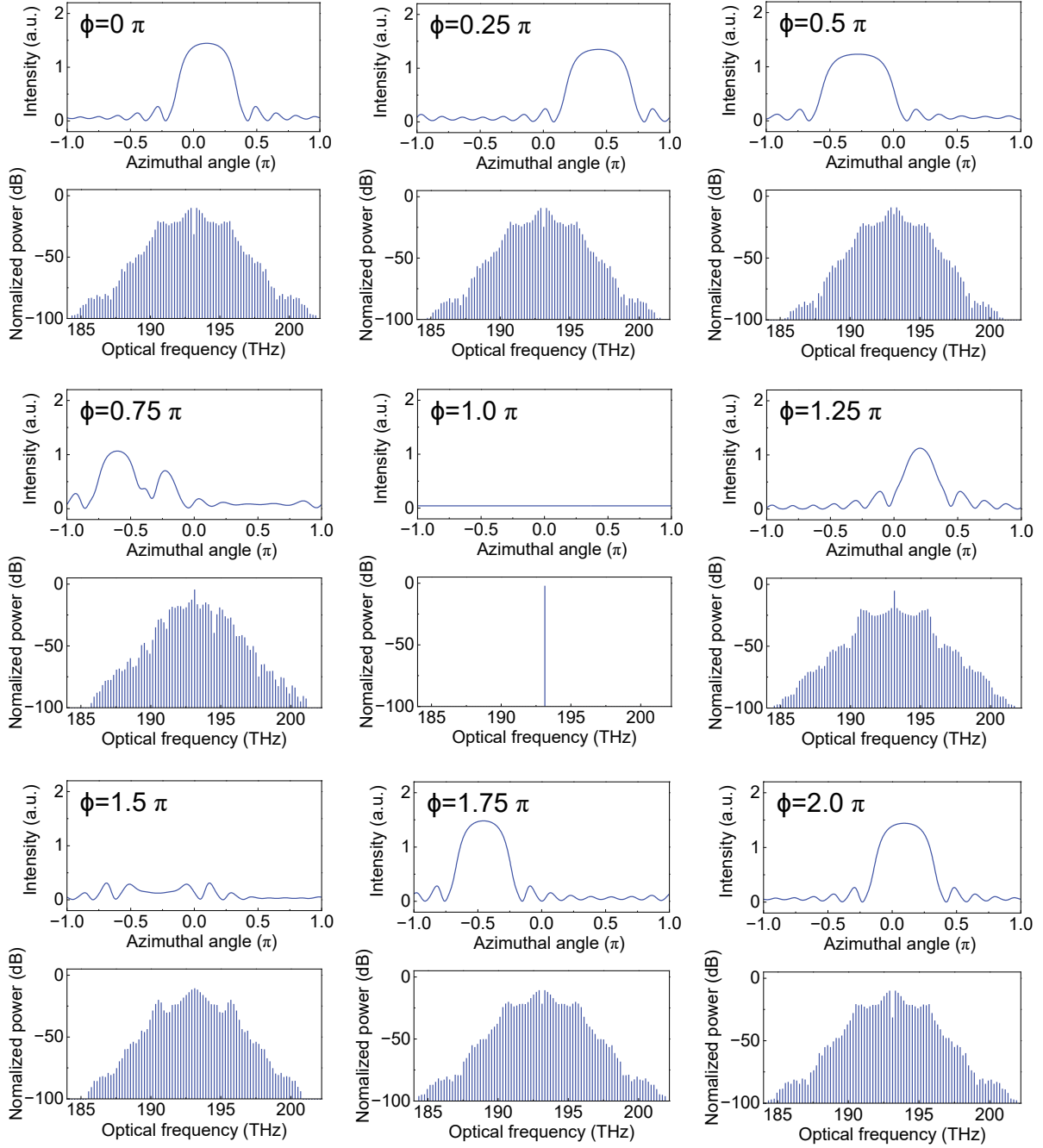


Fig. S2. Simulated intensity profile and comb spectra as ϕ varies. In each panels, the upper and lower sub-panels represent the intracavity intensities and the corresponding spectra, respectively. Those comb spectra corresponds to the simulated CE (blue line) in Fig. 1e.

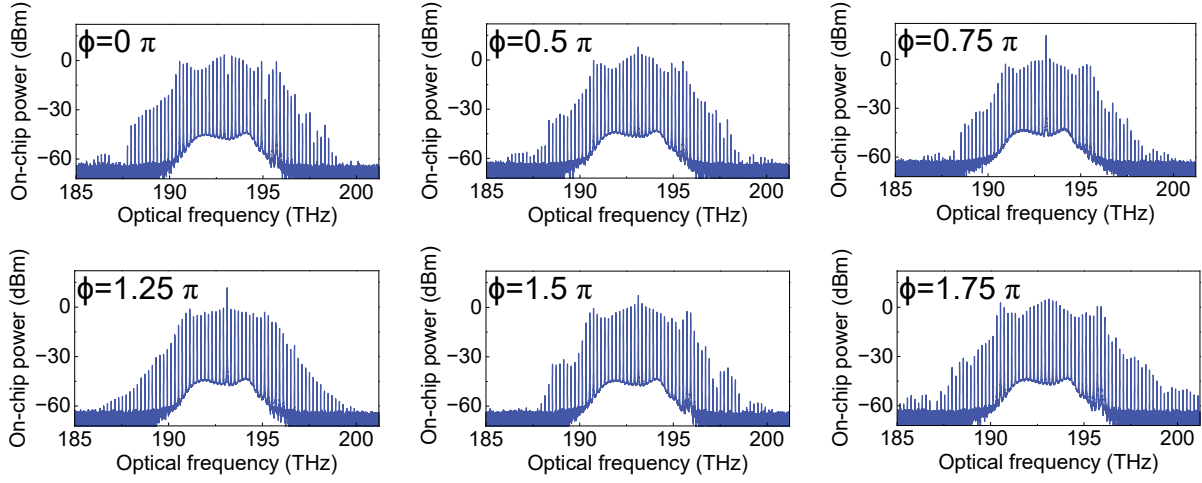


Fig. S3. Measured spectra of the backward combs in the case with different ϕ . Those comb spectra corresponds to the measured CE (circles) in Fig. 1e.

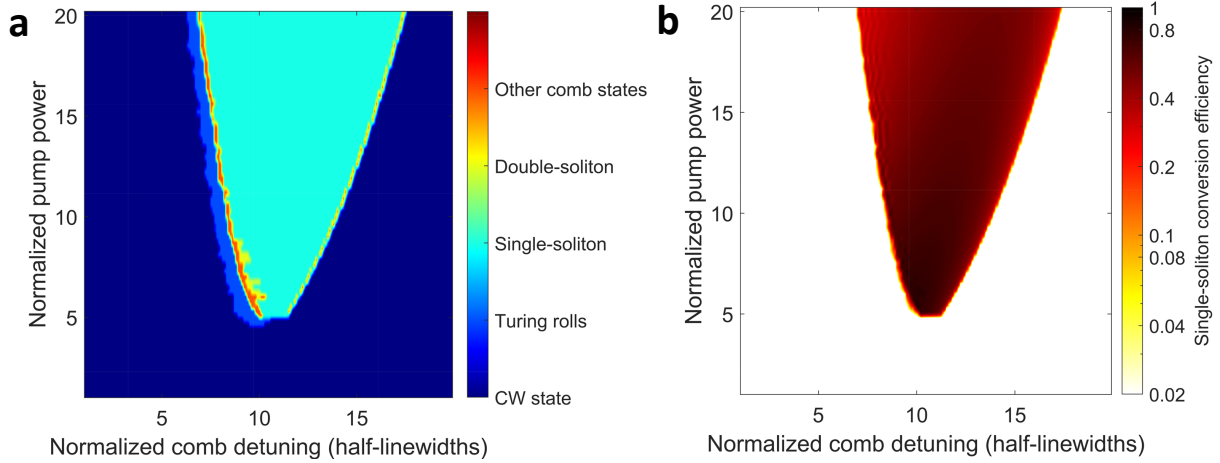


Fig. S4. Simulated stability chart showing the comb states (a) and CE (b) as functions of detuning and normalized pump power .

which is similar to that in [1]. Compared with the existence map in [1], we observe two advantageous features. First, our nanophotonic circuit mainly supports the CW state and the soliton state, namely there is nearly no other competing comb states and the single-soliton states exists over a broad parameter space. Operationally, we can stably access the single-soliton state by increasing the detuning. Second, our nanophotonic circuit can maintain a high conversion efficiency over a wide range of pump power and detuning , which make it more robust against fabrication uncertainties and operational parameters. Benefiting from these two features, our design is more suitable for mass production of high-efficiency microcombs and more effective

for practical applications.

S5. Soliton generation in the PhCR

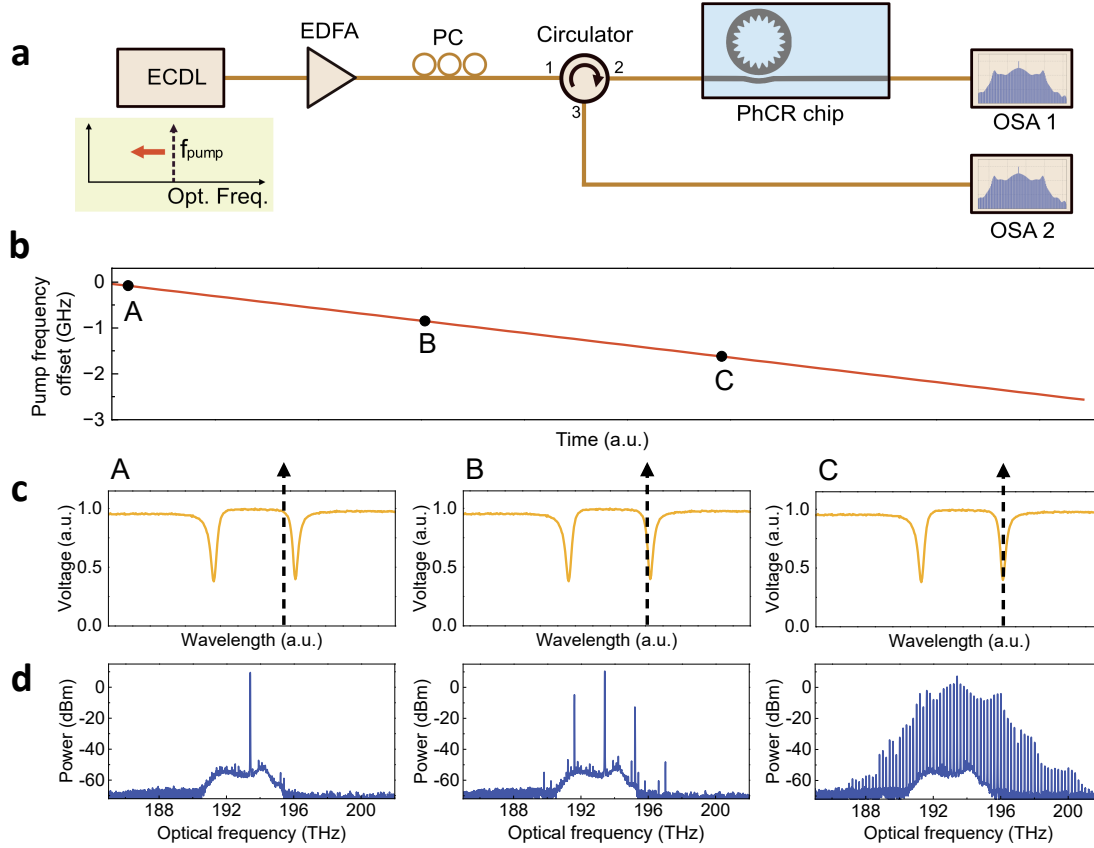


Fig. S5. Experimental initiation of the single-soliton state **a**, Experiment setup, ECDL: external cavity diode laser, PC: polarization controller, OSA: optical spectrum analyzer. **b**, Measured laser frequency change as a function of time. **c**, Illustrations of the split mode and pump laser wavelength as the detuning changes. **d**, Measured comb spectra in the backward direction (OSA 2) as detuning increases.

We generate soliton by pumping the lower-frequency resonance of the split mode and slowly increasing the detuning. Figure S5 shows the process of initiating single soliton in a PhCR without the waveguide reflector. In the case with the waveguide reflector, soliton generation is more straightforward since there is no interference from the thermal triangle of the higher-frequency resonance (see Fig. 2d). Figure S5a shows our experimental setup. The pump is derived from an external cavity diode laser (ECDL, Toptica CTL1550) and coupled to an EDFA to boost the pump power. We use a polarization controller (PC) to vary the pump polarization and a fiber circulator to extract the backward-propagating comb, respectively. The forward and

backward combs are monitored by two optical spectrum analyzers (OSAs). Although it is not stringently required for soliton initiation, we still use a temperature controller to keep a constant device temperature, so that it can maintain the soliton state for a few hours.

To generate soliton, we first tune the laser frequency to the lower-frequency side of the pump mode and then finely increase the detuning. Figure S5b illustrates the pump frequency change as a function of time. Operationally, we control the pump frequency by manually tuning the piezoelectric voltage of the ECDL. Figure S5c shows the sketched split mode (yellow trace) and the laser wavelength (black dashed arrow) as we increase the detuning, while Fig. S5d presents the corresponding spectra of backward combs. Fig. S5 shows that we can generate single-soliton microcomb by slowly increasing the detuning by 2 GHz, which requires simpler setup and easier operation compared with the case with anomalous dispersion cavities [1].

S6. Characterization of repetition rate noise and comb intensity noise

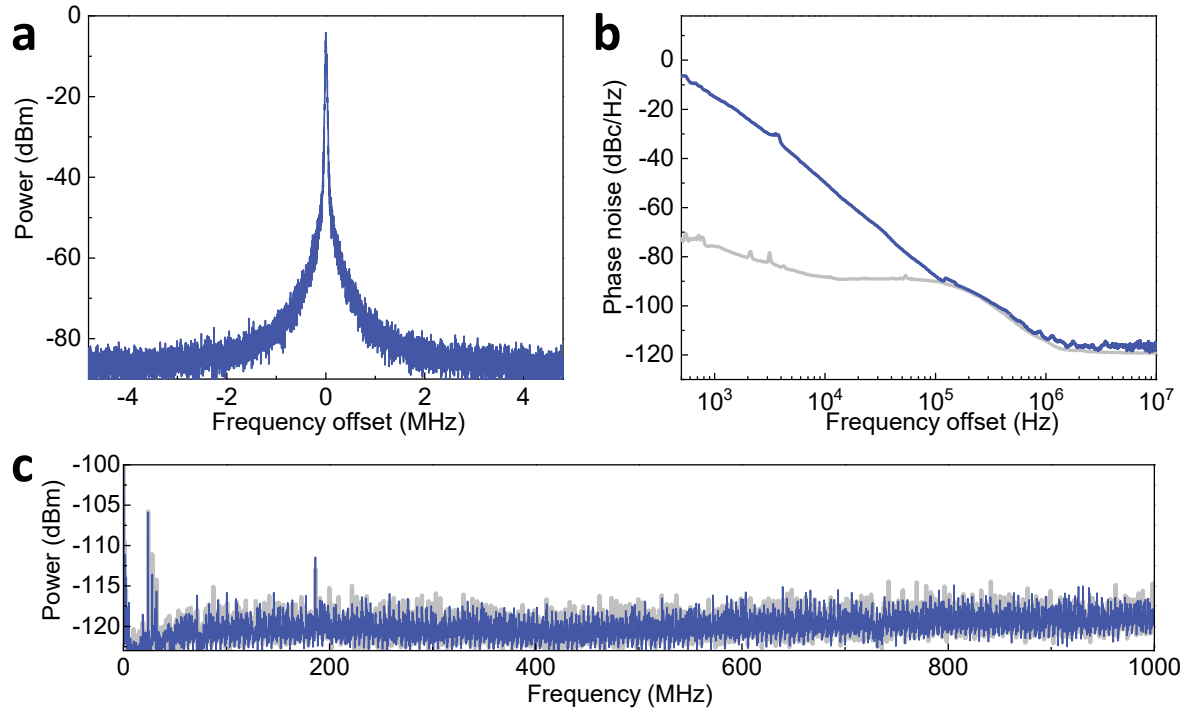


Fig. S6. Characterization of repetition rate noise and comb intensity noise. **a**, Electrical spectrum of the repetition rate after frequency down-conversion and measurement with a resolution bandwidth of 1 kHz. **b**, Measured phase noise of the repetition rate (blue). The gray trace represents the noise floor of our phase-noise measurement system. **c**, Comb intensity noise (blue) measured by sending 2 mW of comb power to a photodiode, the gray trace represents the case without the comb.

We characterize the repetition rate noise after frequency down-conversion using electro-optic modulation. Details of this method can be found in Fig. 3 of [2]. In our experiment, we down-convert the repetition rate from 199.2 GHz to 104 MHz for measurement with a low-frequency electrical spectrum analyzer (ESA). Figure S6a shows the electrical spectrum of the repetition rate measured with a resolution bandwidth of 1 kHz. The narrow peak in the spectrum indicates highly-coherent comb lines generated by our nanophotonic circuit. The corresponding phase noise is presented by blue trace in Fig. S6b, while the measurement noise floor is shown by the gray trace. Above 100 kHz frequency offset, the measured phase noise is limited by our phase-noise measurement system. Furthermore, we measure the comb intensity noise by filtering out the pump and sending 2 mw of comb power to a photodiode; see Fig. S6c. The background of ESA is shown by the gray trace, in which case there is no optical signal at the input of the photodiode. Despite a few spurs induced by the ESA, the microcomb generated by our nanophotonic circuit exhibits intensity noise below -120 dBm from DC to 1000 MHz, indicating a low-noise and stable soliton microcomb that is free of breathing state.

S7. Soliton generation with a DFB laser diode

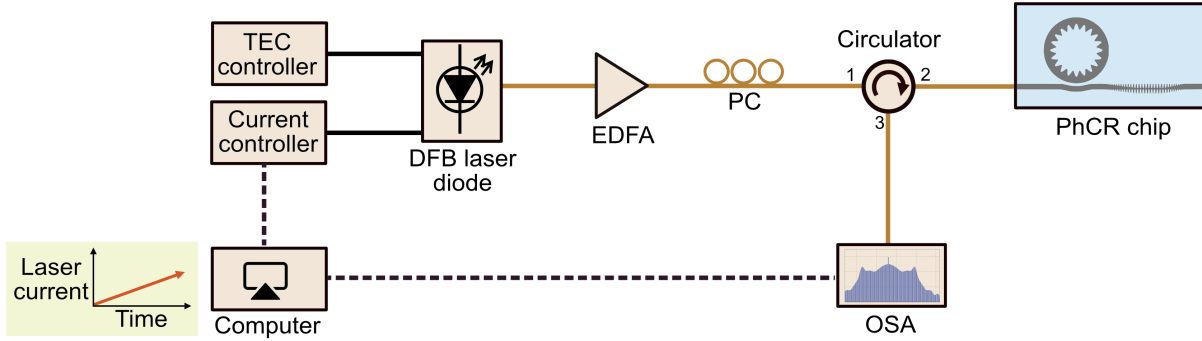


Fig. S7. Experimental setup for soliton generation with a DFB laser diode. TEC: thermo-electric cooler, PC: polarization controller.

The robustness of our design is highlighted by soliton generation with a standard, butterfly-packaged DFB laser (Fig. 4d). Figure S7 shows the experimental setup, which is similar to that in Fig. S4a, except that we replace the ECDL with a commercial telecom laser diode (Optilab DFB-1551-SM-10) and use a PhCR with the waveguide reflector. While the device we use only needs 40 mW on-chip pump power for soliton generation, we still use an EDFA due to the fiber-to-chip coupling loss (4 dB/facet) and low output power (10 mW) of our DFB laser. Operationally, we first align the laser frequency with the lower-frequency resonance of the split mode by tuning the TEC controller. Then we finely increase the detuning through the laser current controller. The corresponding frequency tuning rates are -11.3 GHz/K and -0.9 GHz/mA, respectively. To draw the colormap in Fig. 4d, we program the laser current

controller and OSA, so that we can increase the laser current with a step size of 0.05 mA and simultaneously record the comb spectra during the laser current scan. A laser current increase of 2 mA, corresponding to a detuning increase of 1.8 GHz, is sufficient to initiate the single soliton state in our nanophotonic circuit. This experiment fully demonstrates the compatibility of our nanophotonic circuit with the chip-scale DFB lasers.

References

- [1] Óskar B. Helgason *et al.* Surpassing the nonlinear conversion efficiency of soliton micro-combs. *Nature Photonics* **17**, 992–999 (2023).
- [2] Zang, J., Briles, T. C., Morgan, J. S., Beling, A. & Papp, S. Millimeter wave frequency synthesizer based on integrated photonics. In *2020 International Topical Meeting on Microwave Photonics (MWP)*, 101–104 (2020).